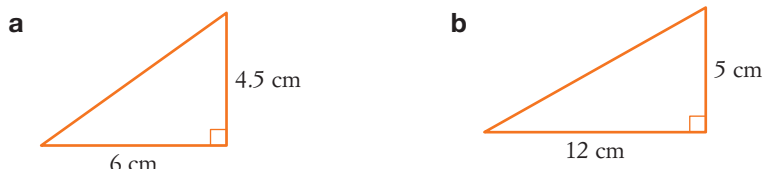


WORKSHEET

Pythagorean triads

- 1 Construct each triangle accurately and measure the length of the **hypotenuse**. Verify that they satisfy Pythagoras' theorem.



- 2 Any set of three numbers that satisfy Pythagoras' theorem is called a **Pythagorean triad**. So (4.5, 6, 7.5) and (5, 12, 13) are examples of Pythagorean triads — the exact lengths of the sides of a right-angled triangles. The first 62 triads are listed below. Select any set and show that it satisfies Pythagoras' theorem.

PYTHAGOREAN TRIADS (exact sides of a right-angled triangle)

(3, 4, 5) (5, 12, 13) (6, 8, 10) (7, 24, 25) (8, 15, 17) (9, 12, 15) (9, 40, 41) (10, 24, 26)
 (11, 60, 61) (12, 16, 20) (12, 35, 37) (13, 84, 85) (14, 48, 50) (15, 20, 25) (15, 36, 39)
 (16, 30, 34) (16, 63, 65) (18, 24, 30) (18, 80, 82) (20, 21, 29) (20, 48, 52) (20, 99, 101)
 (21, 28, 35) (21, 72, 75) (24, 32, 40) (24, 45, 51) (24, 70, 74) (25, 60, 65) (27, 36, 45)
 (28, 45, 53) (28, 96, 100) (30, 40, 50) (30, 72, 78) (32, 60, 68) (33, 44, 55) (33, 56, 65)
 (35, 84, 91) (36, 48, 60) (36, 77, 85) (39, 52, 65) (39, 80, 89) (40, 42, 58) (40, 75, 85)
 (40, 96, 104) (42, 56, 70) (45, 60, 75) (48, 55, 73) (48, 64, 80) (48, 90, 102) (51, 68, 85)
 (54, 72, 90) (56, 90, 106) (57, 76, 95) (60, 63, 87) (60, 80, 100) (60, 91, 109) (63, 84, 105)
 (65, 72, 97) (66, 88, 110) (69, 92, 115) (72, 96, 120) (80, 84, 116)

- 3 Multiplying (or dividing) each number in a triad by the same value gives another triad.
 For example:

$$(3, 4, 5) \times 2 \text{ gives } (6, 8, 10)$$

$$\times 3 \text{ gives } (9, 12, 15)$$

$$\div 2 \text{ gives } (1.5, 2, 2.5)$$

For each triad below, create another triad by multiplying by a constant value, then check that it satisfies Pythagoras' theorem.

a (7, 24, 25) _____ **b** (8, 15, 17) _____

- 4 In the 16th century BCE, the Babylonians invented a formula for generating Pythagorean triads (a , b , c), one thousand years before Pythagoras was born!

If p and q are any two whole numbers and $p > q$, then:

$$a = p^2 - q^2 \qquad b = 2pq \qquad c = p^2 + q^2$$

Substituting $p = 3$ and $q = 2$ gives (5, 12, 13).

Find the Pythagorean triads (a, b, c) if:

a $p = 4, q = 3$ _____

b $p = 3, q = 1$ _____

c $p = 6, q = 4$ _____

d p and q are two numbers you choose. _____

(Advanced students: notice that a^2 is a square multiple of the sum of b and c .)

5 Pythagoras himself invented this formula for generating Pythagorean triads.

If n is any odd number, then: $a = n, b = \frac{n^2 - 1}{2}, c = \frac{n^2 + 1}{2}$

Substituting $n = 3$ gives $(3, 4, 5)$.

Find the Pythagorean triads (a, b, c) if:

a $n = 7$ _____

b $n = 11$ _____

c n is any number you choose. What pattern do you notice about b and c ?

6 A third formula allows n to be odd or even: $a = 2n + 1, b = 2n^2 + 2n, c = 2n^2 + 2n + 1$

a Substitute $n = 4$ to find (a, b, c) . _____

b Substitute another value of n . _____

c What do you notice about a^2 and $b + c$? _____

d What do you notice about $c - b$? _____

Note: These three formulas do not necessarily generate all possible Pythagorean triads.

Challenge: An ancient Chinese puzzle

A bamboo stick which is 32 cubits high is snapped by the wind and now its tip touches the ground 16 cubits from its base. Where did the bamboo snap?

Answers

- 1 a** 7.5 cm **b** 13 cm
- 4 a** (7, 24, 25) **b** (8, 6, 10) **c** (20, 48, 52)
- 5 a** (7, 24, 25) **b** (11, 60, 61) **c** b and c differ by 1
- 6 a** (9, 40, 41) **c** $a^2 = b + c$ **d** $c - b = 1$

Challenge 12 cubits above ground level